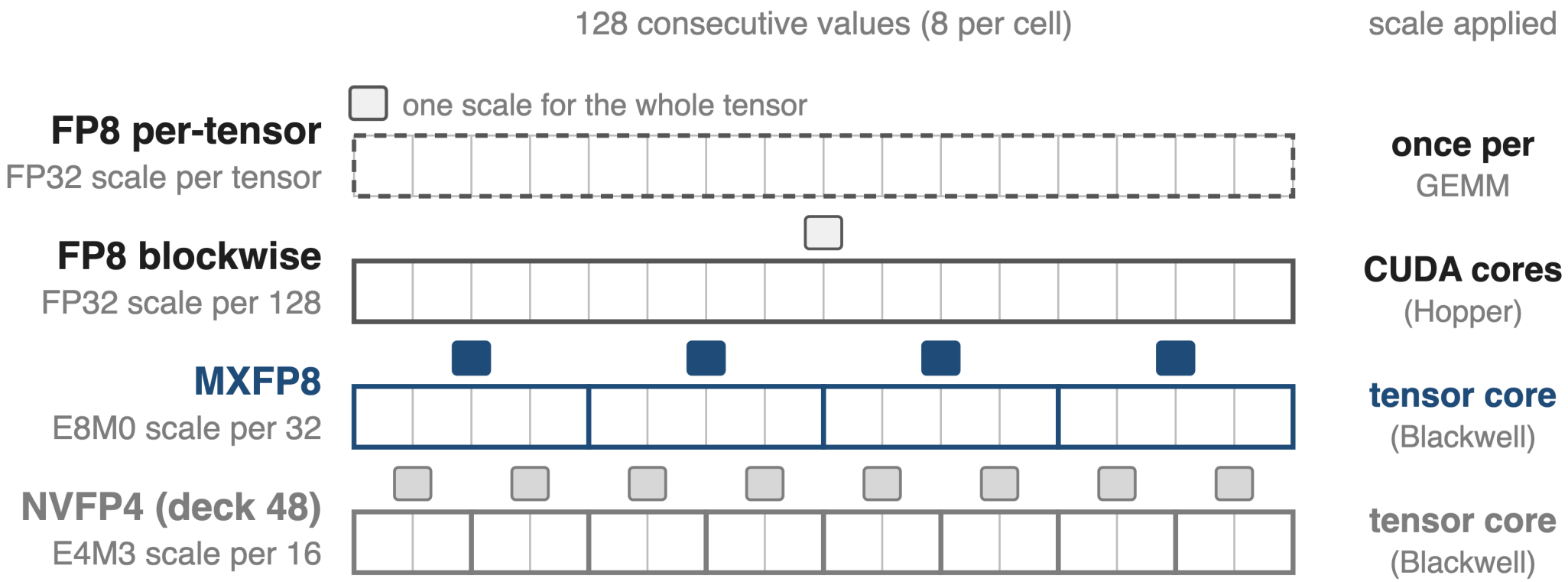
MXFP8 Micro-Scaling Training

FP8 with one hardware scale per 32 values

NVIDIA · Mishra, Stosic, Layton, Micikevicius (arXiv 2506.08027) · Transformer Engine

Explainer deck · Industrial Pretraining Architecture & Infra Tech Landscape (2026)

Blackwell moves FP8 scales into the tensor core



Megatron-Core's default FP8 recipe on Blackwell: DeepSeek-V3 trains at 1,048 TFLOPS/GPU on GB200

On Hopper, finer FP8 scales cost GEMM speed

1.41×

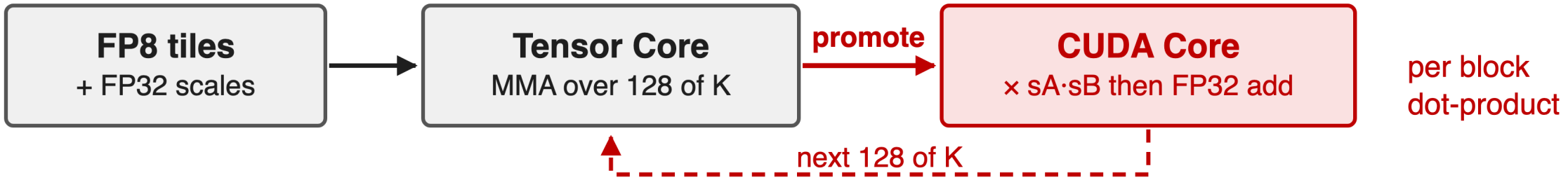
blockwise FP8 GEMM speedup over BF16 on H200, below per-tensor current scaling (1.57×)

- The accurate recipe is the slow one: block scales are **software-emulated** on Hopper

Transformer Engine docs, "GEMM Speedups Across Precisions": 5B model on H200 NVL, autocast (deck 46)

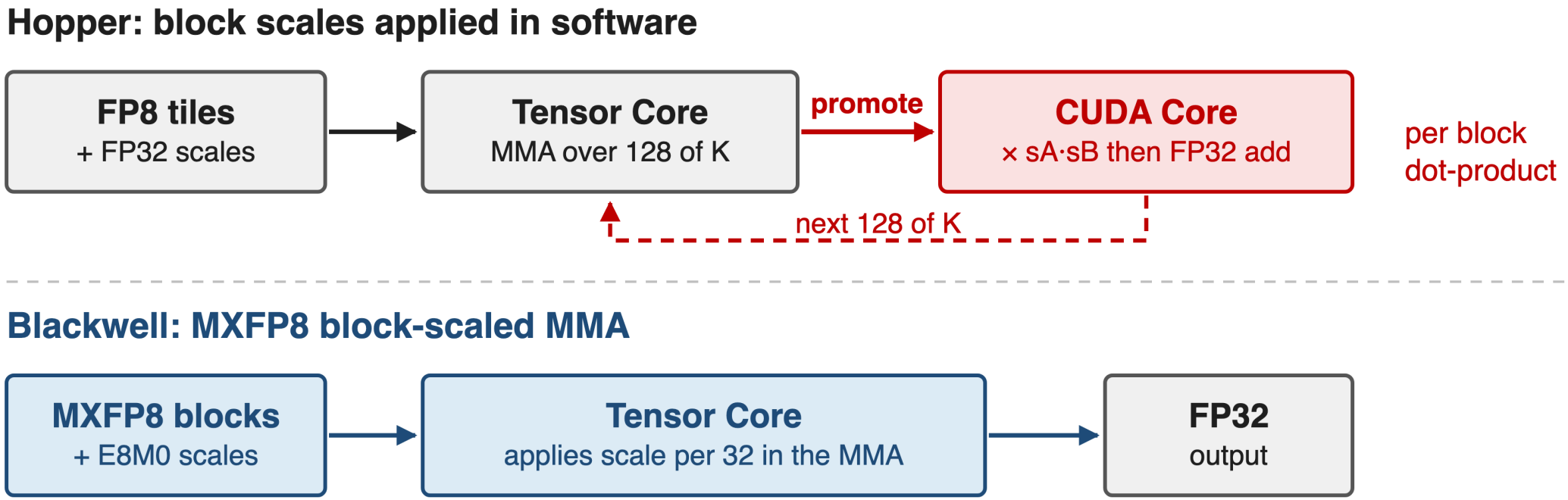
Software block scales detour through CUDA cores

Hopper: block scales applied in software



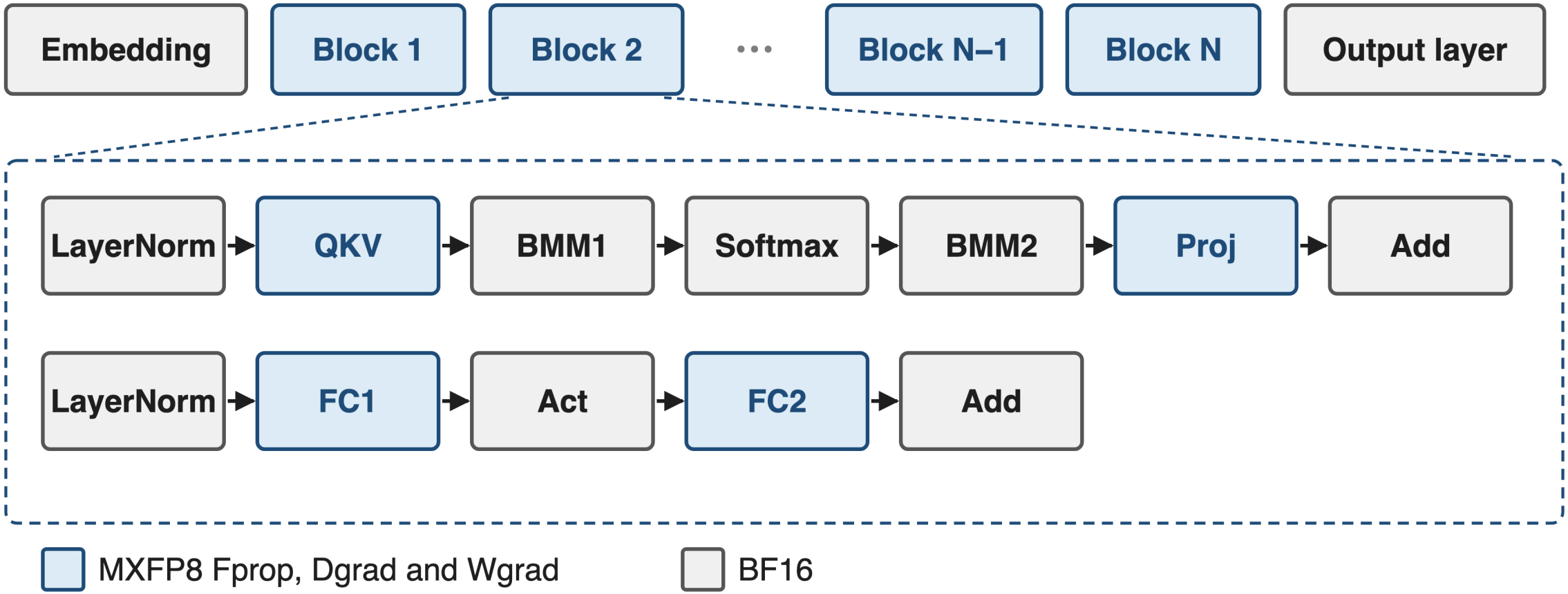
Per-tensor FP8 skips the detour, but keeps E5M2 gradients and leaves some blocks in BF16

Insight: let the tensor core apply the scale



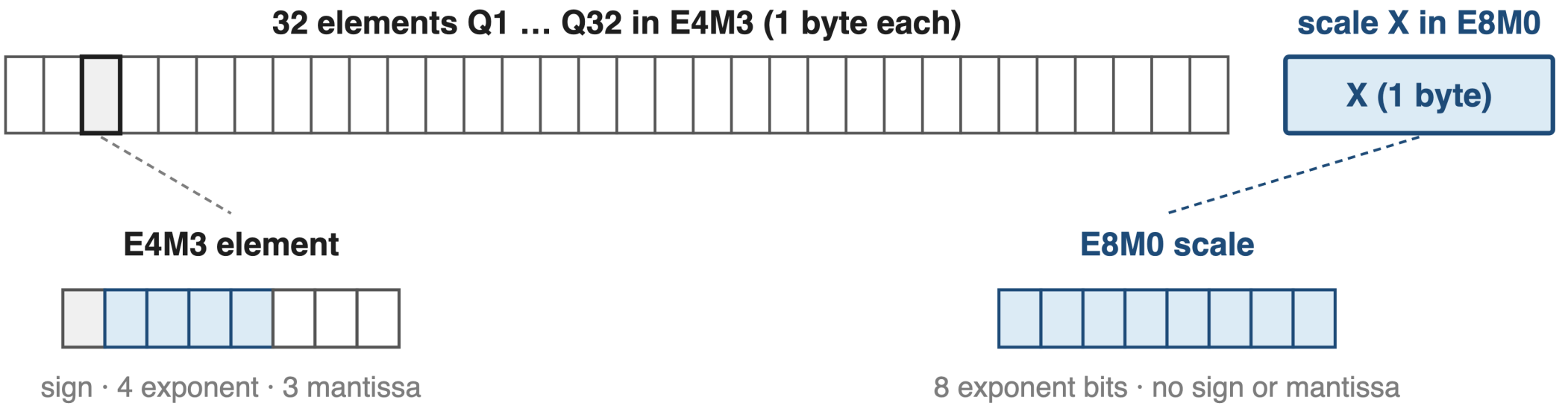
With scaling in hardware, small blocks cost no speed: shrink them to 32, then fix the numerics

MXFP8 quantizes every linear layer of every block



❶ the MX block format · ❷ E4M3 for all tensors · ❸ round the scale up

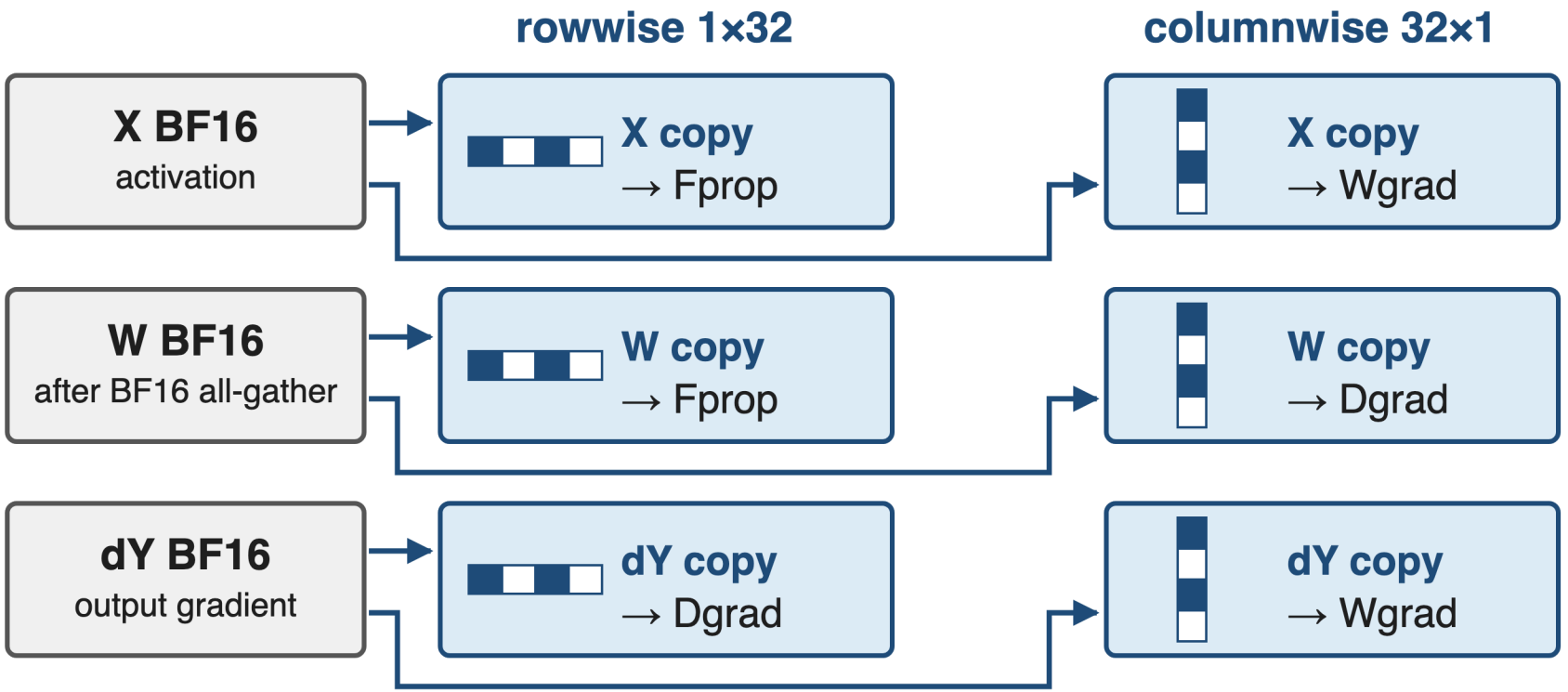
❶ 32 values share one power-of-two scale byte



$$\text{value} = Q_i \times 2^{X-127}$$

8.25 bits per value: the same overhead as 1×128 FP32 scales, 4× finer (derived)

❶ Each operand needs row and column copies

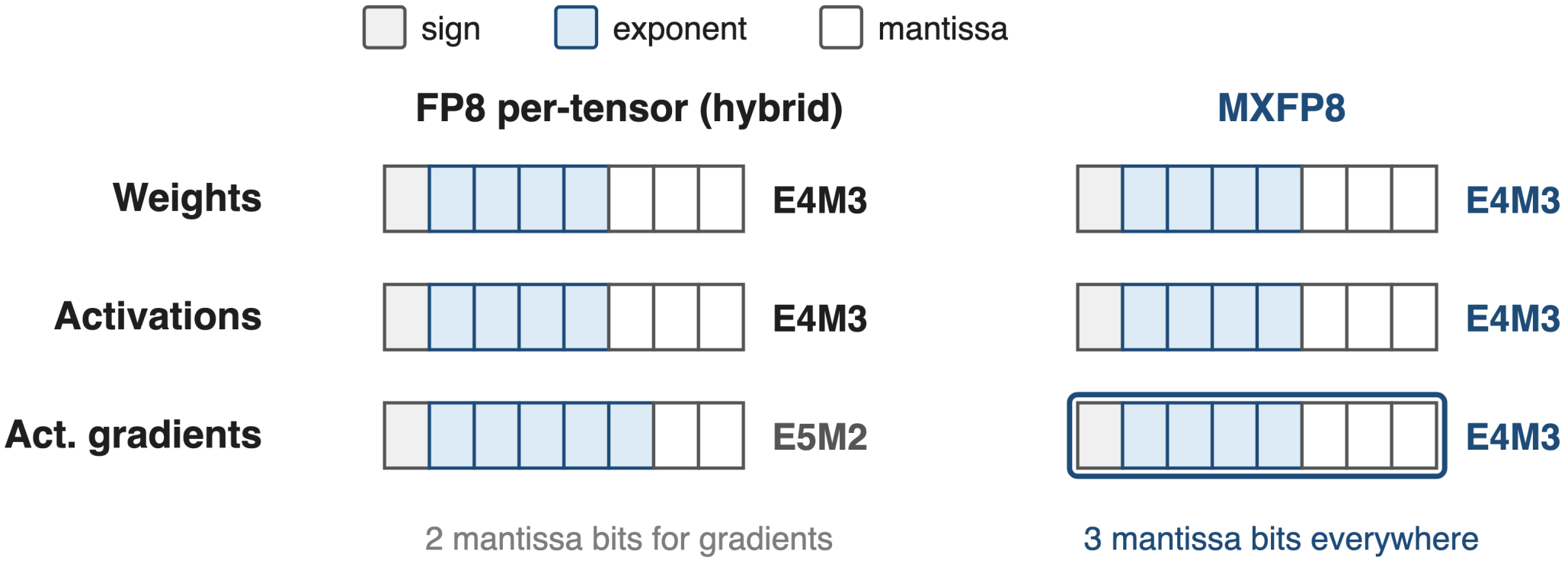


quantized separately from BF16

no transpose needed on Blackwell

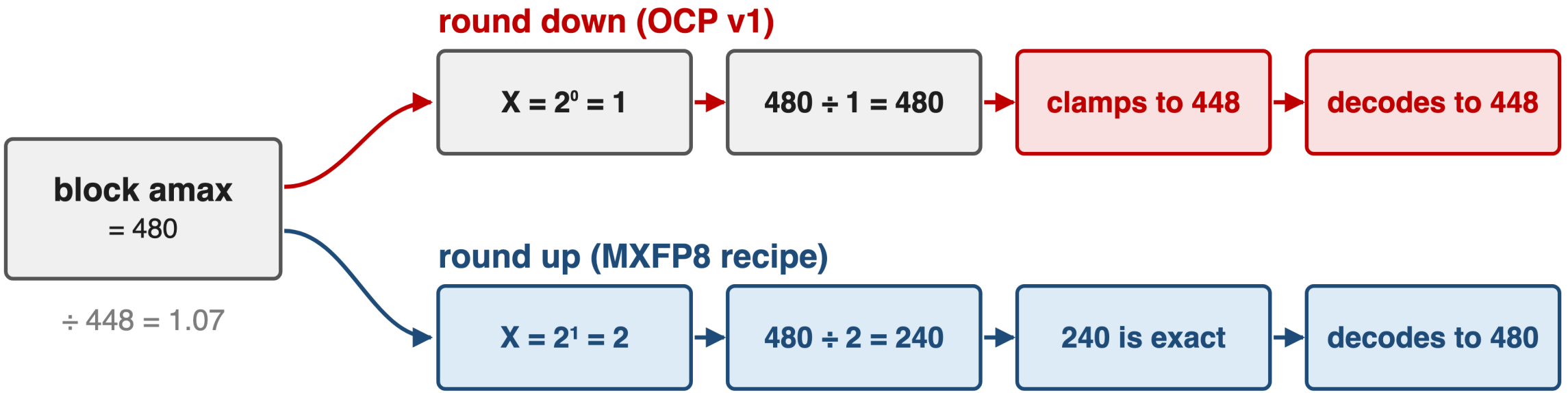
Copies cannot be derived from each other, so MXFP8 weights are all-gathered in BF16

❷ E4M3 everywhere, gradients included



Within 32 values, E4M3's 17.8 binades cover the range; the extra mantissa bit matters

❸ Round the scale up, not down



OCP v1's round-down overflows the block max; round-up removed the 843M training issues

MXFP8 tracks BF16 within 0.5% over 15T tokens

< 0.50%

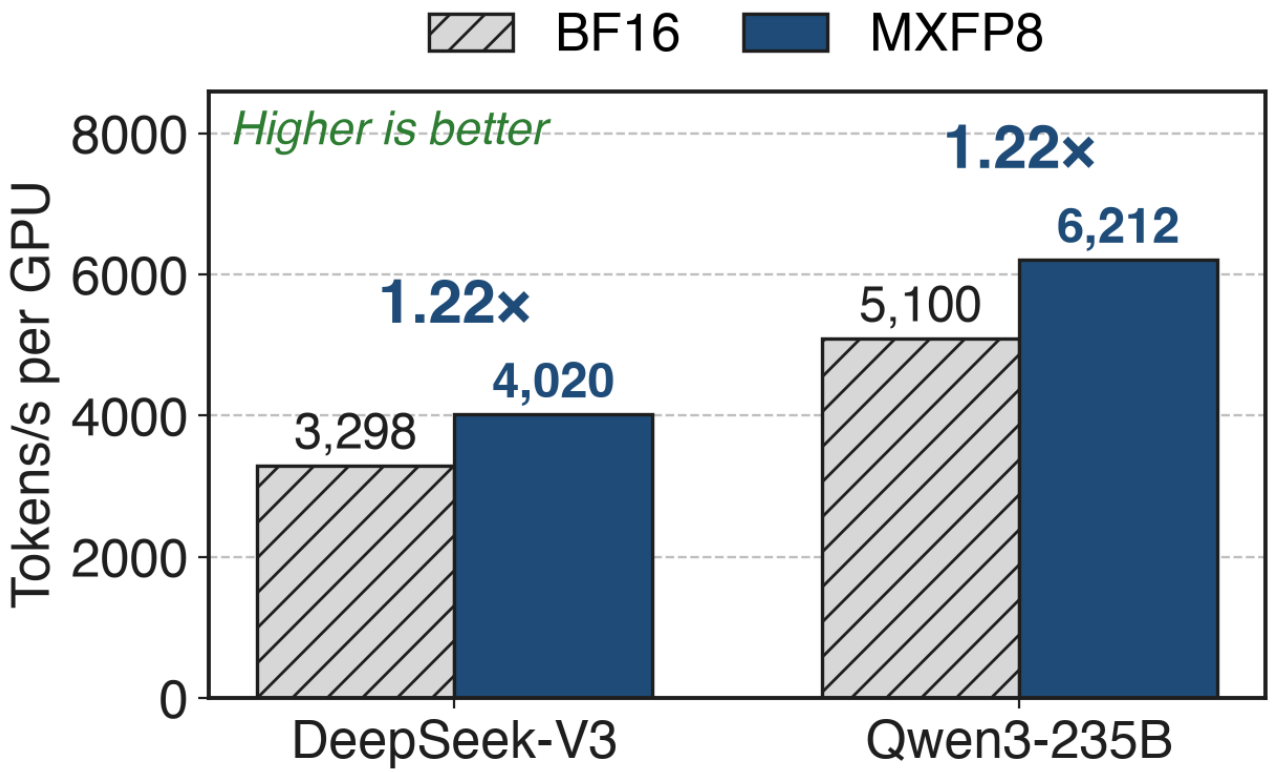
validation-perplexity gap vs BF16 throughout pre-training, 8B model, 15T tokens

- 16B MoE (2.5B active), 1T tokens: final loss within **0.1%**

Mishra et al., arXiv 2506.08027, Sec. 3.1 and App. A.4 (MX numerics simulated on 3,072 Hopper GPUs)

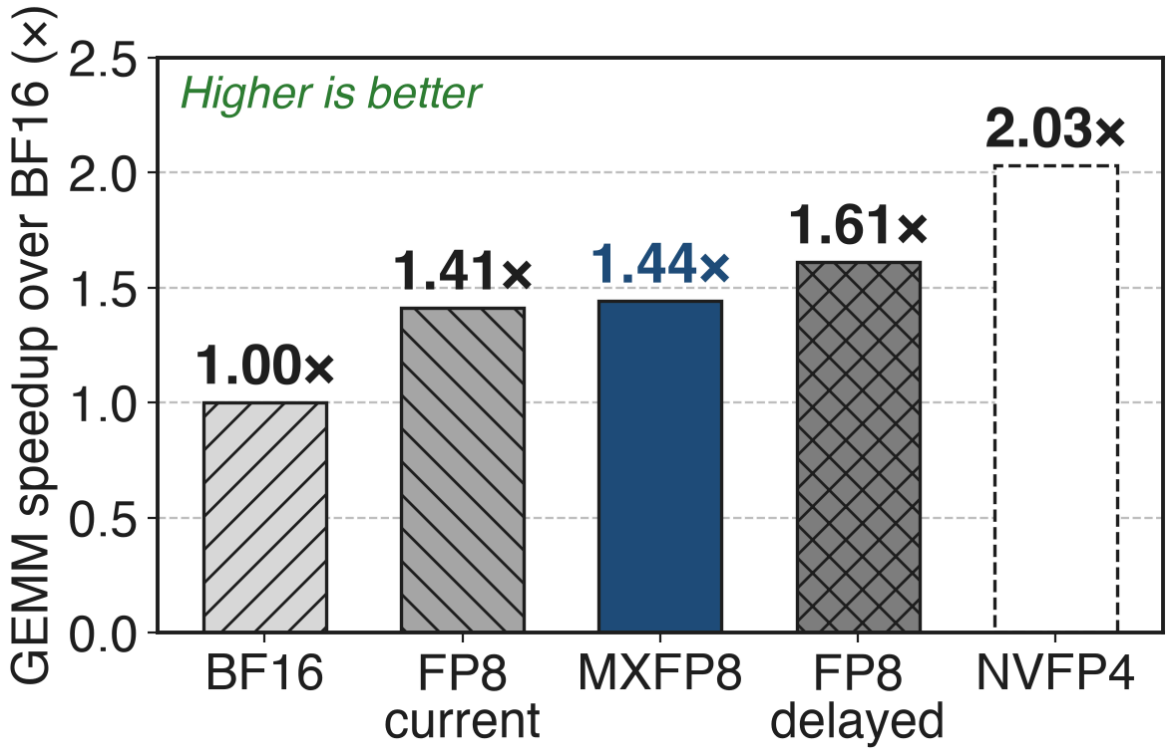
MXFP8 trains DeepSeek-V3 1.22× faster than BF16

- 256 GB200 GPUs, 4k sequence
- Each precision with its own best parallel config
- GB300 in MXFP8: **1,233** TFLOPS/GPU



On B300, MXFP8 GEMMs match per-tensor speed

- 5B model, quantization included
- Blackwell: MXFP8 **1.44×** vs current **1.41×**
- Hopper: blockwise **1.41×** vs current **1.57×**



Delayed scaling is still fastest (1.61×), but it trades precision for speed

The price: two copies, BF16 all-gather, simulation

- Two quantized copies per tensor: memory and quantize kernels
- Parameters all-gathered in BF16; shapes divisible by 32
- Published accuracy runs were simulated on Hopper

Blackwell only (SM 10.x); NVFP4 (deck 48) halves the bits again

Summary

- Hopper's fine FP8 scales live in software: **1.41×** vs **1.57×** per-tensor
- **Insight**: tensor cores apply a power-of-two scale per 32 values; E4M3 everywhere, scale rounded up
- **< 0.50%** perplexity gap at 8B / 15T; **1.22×** end to end on GB200

https://arxiv.org/abs/2506.08027